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Urbana, IL 61801, United States

PROFESSIONAL EXPERIENCES

Postdoctoral Researcher, University of Illinois Urbana-Champaign (UIUC), Illinois

Jan 2026 - present

Goals: Study high-entropy alloy thin films with radiation-induced defect architectures and metastable microstructures.

Roles: Principal experimental contributor to thin film deposition and characterization (Advisor: *Prof. Marie Charpagne*)

- Synthesized wafer-scale (500nm) Ni-based alloy films via DC magnetron sputtering, optimizing deposition parameters.
- Conducted defect and grain-structure analysis through hands-on focused ion beam (FIB) lamella preparation and operation of a Talos F200X G2 scanning transmission electron microscope (S/TEM).

Postdoctoral Researcher, Lawrence Livermore National Laboratory (LLNL), California

June 2022 – Dec 2025

► **Joint project with the group of Prof. Steven Chu (1997 Nobel Laureate in Physics and former U.S. Secretary of Energy) at Stanford University**, Laboratory Directed Research Development (LDRD): “Radiation Defect Engineering in Two-Dimensional Membranes (hexagonal boron nitride, hBN)” *June 2022 – January 2025*.

Goals: Develop scalable ultrathin hBN films with tailored defect architectures for Li-metal battery applications.

Roles: Led experimental efforts and served as primary liaison to the Chu group (Advisor: *Dr. Sergei Kucheyev*)

- Conducted systematic studies on radiation damage and defect formation in hBN polymorphs using 4MV ion accelerator at LLNL (radiation damage to $\sim 1\text{dpa/SRIM}$), mapped damage buildup and phase transformations (*ACS Applied Materials & Interfaces* 2026, *Scripta Materialia* 2025).
- Synthesized wafer-scale, 500nm-hBN films via PVD-DC/RF magnetron sputtering, optimizing key parameters.
- Performed plasma diagnostics (EQP-MS, OES, Langmuir probe, RGA) to analyze B/N/Ar ion energy distributions.
- Characterized films using Raman, FTIR, XAS (ALS/Berkeley National Lab beamline 8.0.1), TEM analysis, identified π^*/σ^* , sp^2 -bonding defect signatures in collaboration with LLNL computational team (*J. Phys. Chem. Lett.*, 2025).
- Teamwork with Stanford on liquid Li metal interaction with sputter-grown h-BN on stainless steel for electrochemical testing, enabling image analysis of Li metal behavior through ex-situ postmortem coin cell analysis (at *Prof. Yi Cui's* lab).
- Authored one first- and corresponding-author publication with Prof. Chu; four additional manuscripts in preparation.
- Collaborated with *Prof. Kun-Yu Lai* (National Central University, Taiwan) on XAS measurements, manuscript submitted.

► **National Ignition Facility (NIF) target fabrication project, 2025**

Focus: Develop boron carbide (B_4C) laser-fusion target capsules and high-entropy, solid solution alloy hohlraum coatings (rare earth Gd + bal. BiTaAuW) to enhance inertial confinement fusion (ICF) energy yield. (PI: *Dr. Sergei Kucheyev*)

Role: Principal experimental contributor to alloy and boron carbide films synthesis process and characterization.

- Optimized surface properties via metrology of macroscopic surface nodular defects; coauthored work with researchers (*Prof. Matthew Wong at UT Dallas*) at Blue Laser Fusion, founded by *Prof. Shuji Nakamura (2014 Nobel Laureate in Physics, UCSB)*, *J. Appl. Physics* (2025).
- Systematic studies on columnar, porosity/densification of films using SEM image analysis and laser confocal microscope as well as residual stress analysis and surface roughness.
- Conducted comprehensive materials characterization (Raman, XRD, XAS, FTIR) confirming amorphous bonding structures. Use of annealing up to 1000 °C to enhance the stability of boron carbide amorphous structures.
- Synthesized 1-30 μm -thick combinatorial high-entropy films using PVD under controlled vacuum conditions; optimized film uniformity and thickness, under diverse conditions; bias voltage for the deposition -50V, oblique angles (0 - 90°).
- Performed Rutherford Backscattering (RBS) and RUMP simulations to evaluate oxygen concentrations (5-10 at.%), and identified oxide phase evolution via XRD.

► **Graduate Researcher, The Pennsylvania State University, Pennsylvania**

August 2018 – June 2022

Focus: Investigation on the microstructural modification of plasma-facing component (tungsten divertor) in edge localized mode (ELMs)-like high heat flux of a tokamak fusion reactor. Conducted ex-situ post-mortem characterization of surface morphology across nano- to microscale dimensions, identified capillary/porous resolidified layers, grain boundary voids, and hardening, providing critical insights into tungsten degradation under fusion environments (npj Mater. Degrad., 2020).
• TEM microphotography studies of grain structures, pore formation, and dislocation environments (J. of Nucl. Mater., 2022, Fusion Eng. Des., 2024).

• Investigated ultrafast laser-focused surface modifications on tungsten and tungsten carbide in air, with a focus on surface oxidation and laser-induced nanopatterning (Fusion Sci. Technol., 2025).

• Post-PhD work (Jan 2022-June 2022): Continued research on tungsten-based materials and fusion-relevant surface modification while completing doctoral research and presenting work at the TMS Annual Meeting.

► **Intern, National Research Foundation of Korea, Korea**

February 2018 – July 2018

Supported research program administration, coordination, and government research funding for principal investigators.

► **Graduate Research Assistant, POSTECH, Korea**

February 2015 – February 2017

• Experience in nuclear waste glass synthesis via furnace quenching and durability assessment under water corrosion and electron irradiation, employing solid-state NMR interpretation, thermal analysis (DTA, TMA), and Vickers indentation

• Research experience involving project “*Basic properties of the wasteforms containing long-lived and heat-generating radionuclides*” and collaboration with Korean Atomic Energy Research Institute (KAERI).

► **Sergeant, United States Forces Korea, US Army Garrison – Yongsan, Korea**

December 2010 – September 2012

• Completed mandatory military service as a Korean Augmentation to the United States Army (KATUSA), assigned to United States Forces Korea J6 in Administration (42A: Human Resources Specialist) and Logistics (92Y: Supply Specialist).

EDUCATION

Ph.D. in Nuclear Engineering, The Pennsylvania State University, PA, USA

December 2021

Dissertation: “Surface Damage and Microstructure Deformation of Resolidified Tungsten in High Heat Flux Conditions”
Committee members: **Leigh Winfrey** (Chair, Advisor), **Jean Paul Allain** (former U.S. DOE Director of Office of Fusion), **Arthur T. Motta**, **Venkatraman Gopalan**

M.S. in Nuclear Engineering, Pohang University of Science and Technology (POSTECH), Korea

February 2017

B.Eng. in Fine Chemical Engineering and Applied Chemistry, Chungnam National University, Korea

February 2015

PUBLICATIONS

*Peer-Reviewed Journals, *Corresponding author, underscore = First authored*

► **LLNL – Chu group collaboration**

• **Minsuk Seo***, L. Bimo Bayu Aji, Luis A. Zepeda-Luiz, Sang Cheol Kim, Yan-Kai Tzeng, Steven Chu, Sergei O. Kucheyev, “Dynamics of radiation damage buildup in ultrathin hexagonal boron nitride under ion irradiation” *ACS Applied Materials & Interfaces* (2026) <https://doi.org/10.1021/acsami.5c25472>

• **Minsuk Seo***, L. Bimo Bayu Aji, Sang Cheol Kim, Yan-Kai Tzeng, Alexander A. Baker, Finnian M. O’Neill, Steven Chu, Sergei O. Kucheyev, “Damage buildup in multilayer hexagonal boron nitride films under Ar ion bombardment” *Scripta Materialia* 265, 116756 (2025) doi.org/10.1016/j.scriptamat.2025.116756

► **LLNL**

• James B. Merlo, **Minsuk Seo**, Koki Kawasaki, Liam R. Sohngen, L. Bimo Bayu Aji, Swanee J. Shin, Gregory V. Taylor, Sergei O. Kucheyev, “Effects of discharge power for magnetron sputter deposition of boron carbide films on tilted substrates” *Journal of Applied Physics* (2026) <https://doi.org/10.1063/5.0317058>

• Matthew S. Wong, Gregory V. Taylor, **Minsuk Seo**, L. R. Sohngen, James B. Merlo, L. Bimo Bayu Aji, Swanee J. Shin, Shuji Nakamura, Sergei O. Kucheyev “Effects of hydrogen on sputter-deposited boron carbide films” *Journal of Applied Physics* 137(19) 095301 (2025) <https://doi.org/10.1063/5.0272124>

- Wenyu Sun, **Minsuk Seo**, L. Bimo Bayu Aji, Gregory V. Taylor, Alexander A. Baker, Sergei O. Kucheyev, Liwen F. Wan, “First-Principles simulations correlating X-ray absorption spectroscopy features to point defects in h-BN” *The Journal of Physical Chemistry Letters* (2025): 16, 16, 3926-3931 doi.org/10.1021/acs.jpcllett.5c00768
- Koki Kawasaki, James B. Merlo, Stephanie Gonzalez, L. Bimo Bayu Aji, Swanee J. Shin, Gregory V. Taylor, Alison M. Engwall, **Minsuk Seo**, Alex A. Baker, Matthew S. Wong, Finnian M. O’Neill, Sergei O. Kucheyev, “High-rate magnetron sputter deposition of low-stress boron carbide films on tilted substrates” *Journal of Applied Physics* 138(10) 105302 (2025) <https://doi.org/10.1063/5.0283657>
- **Minsuk Seo**, James B. Merlo, Liam R. Sohnngen, L. Bimo Bayu Aji, Jessica Bober, Qi Rong Yang, Gregory V. Taylor, Sergei O. Kucheyev, “Ion bombardment tailoring of stress and tensile strength in sputter deposited boron carbide” (2026, under review)
- James B. Merlo, Jesscia Bober, **Minsuk Seo**, Liam R. Sohnngen, Qi Rong Yang, Gregory V. Taylor, Sergei O. Kucheyev, “Intrinsic stress and stiffness of boron carbide films” (2026, under review)
- Cheng-Da Tu, Chih-Chieh Chang, Wei-Jie Hsu, Yong-Ji Chen, Jen-Inn Chyi, Chien-Ting Wu, Kun-Lin Lin, **Minsuk Seo**, Augustinas Galeckas, Yun-Chorng Chang, Andrej Kuznetsov, “Overcoming p-channel transistor performance asymmetry with ultrahigh conductance two-dimensional hole gas at the boron nitride/silicon interface” (2026, submitted)
- **Penn State**
- **Minsuk Seo***, Shukai Yu, Venkatraman Gopalan, A. Leigh Winfrey, “Observations on macroscopic surface modification and oxidation of tungsten and tungsten carbide in laser focus in air” *Fusion Science and Technology* (2025): 1-6. doi.org/10.1080/15361055.2024.2343972
- **Minsuk Seo***, A. Leigh Winfrey. “Microstructural modification of highly porous resolidified tungsten in heat flux of 46.3GW/m² and focused ion beam” *Fusion Engineering and Design* (2024): 114206. doi.org/10.1016/j.jnucmat.2022.153725
- **Minsuk Seo***, Ke Wang, John R. Echols, A. Leigh Winfrey. “Microstructure deformation and near-pore environment of resolidified tungsten in high heat flux conditions” *Journal of Nuclear Materials* (2022): 153725. doi.org/10.1016/j.jnucmat.2022.153725
- **Minsuk Seo***, “Assessment of the maximum temperature of simplified high heat radioactive glass in interim storage and deep geological disposal.” *Nuclear Technology* 207 (2021): 1902-1912. doi.org/10.1080/00295450.2020.1860614
- **Minsuk Seo***, John R. Echols, A. Leigh Winfrey. “Morphological and nanomechanical changes in tungsten in high heat flux conditions.” *npj Materials Degradation* 4.1 (2020): 1-8. [[Phys.org](https://www.nature.com)] [[PennState](https://www.pennstate.edu)] doi.org/10.1038/s41529-020-00135-4
- **Manuscript in preparation**
- **Minsuk Seo**, L. Bimo Bayu Aji, Gregory V. Taylor, Finnian M. O’Neill, Alex A. Baker, Sang Cheol Kim, Yan-Kai Tzeng, Alison Engwall, Steven Chu, Sergei O. Kucheyev, “Radiation damage and amorphization in ultrathin hexagonal boron nitride under 3MeV ion irradiation” (2025)
- **Minsuk Seo**, Finnian M. O’Neill, G.V. Taylor, A.A. Baker, L. Bimo Bayu Aji, Sang Cheol Kim, Yan-Kai Tzeng, Alison Engwall, Steven Chu, Sergei O. Kucheyev, Toward scalable plasma-assisted synthesis of ultrathin hexagonal boron nitride films” (2025)
- **Minsuk Seo**, L. Bimo Bayu Aji, Gregory V. Taylor, Sang Cheol Kim, Finnian O’Neill, Alex A. Baker, Yan-Kai Tzeng, Steven Chu, Sergei O. Kucheyev, “Characteristic diversity of ultrathin hexagonal boron nitride films synthesized by RF magnetron sputter” (2024)
- **Minsuk Seo**, Daniel Goodelman, A.M. Engwall, Gregory V. Taylor, Eunjeong Kim, Sergei O. Kucheyev, L. Bimo Bayu Aji, “Combinatorial sputter deposition of thin Gd-TaAuWBi alloy films” (2025)
- **Minsuk Seo**, Daniel Goodelman, Eunjeong Kim, Gregory V. Taylor, L. Bimo Bayu Aji, “Effects of substrate tilt and chamber pressure on magnetron-sputtered Gd coatings” (2025)

TECHNICAL REPORT

Minsuk Seo, 2D-3D Thin Film Processing for Advanced Nuclear Energy Applications, The School of Chemical Engineering, Oklahoma State University, OK, April 9, 2026.

INVITED FACULTY CANDIDATE SEMINARS

underscore = *Speaker*

Minsuk Seo, 2D-3D Thin Film Processing for Advanced Nuclear Energy Applications, The School of Chemical Engineering, Oklahoma State University, OK, April 9, 2026.

Minsuk Seo, 2D-3D Materials for Advanced Nuclear Energy Systems, Department of Physics, University of South Florida, FL, March 9, 2026. https://www.instagram.com/p/DVi_Xd5DhJS/

Minsuk Seo, 2D-3D Materials for Advanced Nuclear Energy Systems, Department of Materials Science and Engineering, University of Texas at Dallas, TX, February 18, 2026.

PROPOSAL PREPARATION

underscore = *PI*

► LLNL

• **M. Seo et al.**, “Phase tuning of boron nitride for battery by irradiation”, LLNL LDRD-exploratory research, \$300k/yr, FY26 (unselected)

Stabilizing SEI in Li metal batteries via CVD-grown hBN and ion irradiation to tailor BN polymorphs.

• **M. Seo et al.**, “Boron nitride phase tuning for battery by irradiation”, LLNL LDRD-lab wide, \$300k/yr, FY26

Tuning BN phase for Li deposition via ion irradiation to suppress dendrites and optimize battery performance. (unselected)

• **M. Seo et al.**, “Fusion irradiated defects in boron nitride for battery applications”, LLNL LDRD-lab wide, \$300k/yr, FY25. Study of radiation effects on hBN using recycled-fusion helium ions for Li battery applications. (unselected)

• **M. Seo et al.**, “Accident-tolerant multilayer coating for nuclear fusion”, LLNL LDRD-lab wide, \$300k/yr, FY25.

Developed tungsten/tungsten carbide multilayered coatings to improve fusion reactor safety under accident conditions. (unselected)

TEACHING AND MENTORSHIP

► Graduate Teaching Assistant, NUCE, The Pennsylvania State University, PA

August 2018 – May 2019

• NUCE 431, Spring 2019 (Core Design Synthesis, *Prof. William Walters*): Facilitated group communication and supported senior-level students in job searching within reactor operations, the energy industry, and processing. Responsibilities included grading weekly assignments, organizing seminars, providing individual feedback, and guiding graduation conferences. Advised an undergraduate student who is now pursuing a PhD under *Prof. Yong-Su Na* (Seoul National University, Korea).

• NUCE 420, Fall 2018 (Radiological Safety, *Prof. Marek Flaska*): Applied strong knowledge of radiation effects in materials and radiolysis. Responsibilities included grading weekly assignments, assisting in lecture preparation, providing individual feedback, and mentoring undergraduate students on coursework and graduate school opportunities.

• Published solo-author paper on thermal simulation of nuclear waste storage and canisters, *Nuclear Technology* (2021).

• Mentored two summer interns on thin film growth, plasma diagnostics, and data analysis.

► Mentorship

Supervised a summer intern on surface metrology, thin film deposition, vacuum system/troubleshooting, career advice.

• Jessica Bober (2025) – currently a Rutgers University student and intern at Princeton Plasma Physics Laboratory.

• Finnian O’Neil (2024) – currently a PhD student at in Nuclear Engineering at University of Wisconsin – Madison.

• Stephanie Gonzalez (2023) – currently a PhD student in Aerospace Engineering at Georgia Tech.

• Rin Choi (2019) – currently a PhD student in Nuclear Engineering at Seoul National University.

PRESENTATIONS

underscore = Speaker

► Oral presentation

D. Goodelman, **M. Seo**, G. Taylor, A. Engwall-Holmes, S. Shin, D. Strozzi, B. Bocklund, J. Chesser, J. Aut, S. Kucheyev, L. B. Bayu Aji, “A combinatorial Approach to develop sputter-deposited lanthanide-containing high entropy alloys for ICF applications, International Conference on Metallurgical Coatings & Thin Films, San Diego, CA, April 21, 2026

M. Seo, F. O’Neill, L. B. Bayu Aji, S. Vangara, S.C. Kim, Y-K Tzeng, CE Kim, Y. Zhou, L. Wan, B. Wang, T.W. Heo, L.A. Zepeda-Ruiz, S. Chu, S. Kucheyev, “Damage buildup in ultrathin boron nitride under ion bombardment”, Materials Research Society Spring 2025, Seattle, WA, April 7-11, 2025

M. Seo, D. Ding, L. Winfrey, “Observations surface modifications on tungsten in laser focus in air”, The Minerals, Metals & Materials Society (TMS) 2025, Las Vegas, NV, March 23-27, 2025

M. Seo, L. B. Bayu Aji, S. Vangara, S.C. Kim, Y-K Tzeng, Y. Zhou, L. Wan, B. Wang, T.W. Heo, CE Kim, L.A. Zepeda-Ruiz, S. Chu, S. Kucheyev, “Radiation defect engineering in hexagonal boron nitride films”, Materials Research Society Fall, Boston, MA, Dec 1-6, 2024

M. Seo, A.L. Winfrey “Highly porous resolidified tungsten microstructures under fusion-like high heat flux and focused ion beam”, American Nuclear Society, Las Vegas, NV, June 16-19, 2024

M. Seo, L. B. Bayu Aji, S.C. Kim, Y-K Tzeng, Y. Zhou, L. Wan, CE Kim, B. Wang, T.W. Heo, L. A. Zepeda-Ruiz, S. Chu, S. O. Kucheyev “Sputter deposition of hexagonal boron nitride films”, International Conference on Metallurgical Coatings & Thin Films, San Diego, CA, May 19-24, 2024

M. Seo, L. B. Bayu Aji, S.C. Kim, Y-K Tzeng, Y. Zhou, L. Wan, CE Kim, B. Wang, T.W. Heo, L. A. Zepeda-Ruiz, S. Chu, S. O. Kucheyev “Plasma-assisted physical vapor deposition of polycrystalline hexagonal boron nitride films”, Materials Research Society Spring, Seattle, WA, April 22-26, 2024

D. Ding, **M. Seo**, L. Winfrey, “Property Degradation and Microstructure Damage of Tungsten Influenced by Helium Plasma”, Technology of Fusion Energy (TOFE), Anaheim, CA, June 12-16, 2022

M. Seo, L. Winfrey, “Microstructure Deformation and Possible Densification of Tungsten in High Heat Flux Conditions”, The Minerals, Metals & Materials Society (TMS) 2022, Anaheim, CA, Feb 27-Mar 3, 2022

M. Seo, “The Maximum Temperature of Heat-generating Glass Waste Canisters in Repository”, American Nuclear Society Decommissioning Environmental Science and Remote Technology 2021, Washington D.C./Virtual, Nov 30-Dec 3, 2021

M. Seo, S. Yu, V. Gopalan, L. Winfrey, “Femtosecond Laser Induced Surface Damages in Tungsten and Tungsten Carbide in High Heat Flux Conditions”, Materials Science and Technology 21, Columbus, OH, Oct 17-20, 2021

M. Seo, S. Yu, V. Gopalan, L. Winfrey, “Morphological Change in Tungsten by Off-normal Angled Femtosecond Laser”, American Nuclear Society Winter, Virtual, Nov 16-19, 2020.

M. Seo, L. Winfrey, “Nanomechanical Change of Tungsten in ELM Conditions”, Materials Science and Technology, Virtual, Nov 2-6, 2020

M. Seo, V. Cintron, A. Cacheris, S. Kosslow, J.R. Echols, L. Winfrey, “High Heat and Particle Damage Effects on Tungsten and Tungsten Carbide”, Materials Research Society Winter, Boston, MA, Dec 1-6, 2019.

M. Seo, J. Heo, H.S. Park, “Glass Waste Forms for Heat-generating Cs⁺ and Sr²⁺ Wastes from Pyro-processing”, The 21st Korean Nuclear Society Spring, Jeju, Korea, May 11-13, 2016. [[pdf](#)]

► Poster

M. Seo, “Plasma-assisted Deposition of Gd and Boron Carbide for NIF”, 2025 Institutional postdoctoral poster symposium, LLNL, Livermore, CA, June 24, 2025

M. Seo, “Sputter Deposition of Hexagonal Boron Nitride Films”, 2024 Institutional postdoctoral poster symposium, LLNL, Livermore, CA, Aug 13, 2024

M. Seo, “Deformation and Resolidification of Tungsten Surface in High Heat Fluxes Using Electrothermal Plasma and Focused Laser, Jeju, Korea, June 6-9, 2023

M. Seo, “Radiation damage buildup in hBN film under ion irradiation”, 2023 Institutional postdoctoral poster symposium, LLNL, Livermore, CA, May 4, 2023

M. Seo, L. Winfrey, “Microstructure Deformation and Possible Densification of Tungsten in High Heat Flux Conditions”, The Minerals, Metals & Materials Society (TMS) 2022, Anaheim, CA, Feb 27-Mar 3, 2022.

M. Seo, J. Heo, H.S. Park, J.H. Choi, “Electron Irradiation Effects on Glass Waste Form for Immobilization of Radioactive Cs⁺ from Pyro-processing”, Proceedings of the Korean Radioactive Waste Conference, Jeju, Korea, Oct 12-14, 2016 [[pdf](#)]

M. Seo, J. Heo, H.S. Park, J.H. Choi, “Glass Waste Forms for Immobilization of Heat-generating Cs⁺ and Sr²⁺ Oxide Wastes from Pyro-processing”, Society of Glass Technology Centenary Conference, Sheffield, UK, Sep 4-8, 2016

TECHNICAL SKILLS

Experimental & synthesis: Ion accelerator at LLNL, Physical vapor deposition (magnetron sputter deposition), High vacuum system, Laser ablation, Electron beam treatment, Tube furnace, Glass quench-synthesis, Glass corrosion test, FEI Helios 600i Dual Beam SEM/FIB, Thermo Scios2 Dual-Beam SEM/FIB, ThermoFisher Scientific Talos F200X G2 Scanning Transmission Electron Microscope (S/TEM)

Characterization: SEM, Raman spectroscopy, FT-IR, EDS, XRD, Helium Pycnometer, Vickers & nanoindentation, DTA, TMA, EQP, AFM, OES, Langmuir probe, RGA, Confocal laser microscopy/KEYENCE, Profilometry

Spectroscopy & Analysis: UV-Vis, XPS, XRF, Solid-state NMR, EPR

Computation tools: ANSYS workbench, MATLAB, LATEX, SRIM, RUMP, Origin, Solidworks, ImageJ, Digital Micrograph, Adobe Illustrator, JADE XRD, EasySpin

PATENT

J. Heo, J-Y. Pyo, C. W. Lee, M. Kim, **M. Seo**, “*Borate glass wasteform to immobilize rare-earth oxides form pyro-processing and manufacturing method thereof*”, 10-20914840, Korea (2020)

AWARDS

- Graduate admission scholarship: Penn State University, United States, 2018.
- Graduate school scholarships for 4 semesters: POSTECH, Korea, 2014.
- 3rd Award: 1st Ulsan Technopark business idea in the renewable energy competition, Korea, 2014.
- Academic excellent scholarship: CNU-Energy MAP, Korea, 2014.
- Academic scholarships for 2 semesters: Chungnam National University, Korea, 2014.
- 2nd Award: 6th Korean Electric Power Corporation Research Institute’s college students’ power technology idea competition, Korea, 2013.

ACTIVITIES

- Journal paper reviewer: Fusion Science and Technology (2020), Journal of Nuclear Materials (2024), Fusion Engineering and Design (2024). IEEE Transactions on Plasma Science (2025, 2026), Fusion Science and Technology (2025)
- Participation certificate: Nuclear Nonproliferation, Safeguard, and the Security in the 21st Century (2-weeks), Brookhaven National Laboratory, NY, June 2019.
- Visitor tour: SKB facilities; spent fuel storage, canister factory, and 500m deep repository, Sweden, Nov 2016.
- Visiting student: Reactor physics experiment program (1-week) at the Kyoto University Critical Assembly, Japan, Dec 2015.
- Public presentation: 1st Gwacheon National Science Museum’s Noble prize essay contest, Korea, 2014.
- Internship: 18th Korean presidential election, Nov – Dec 2012.

PROFESSIONAL REFERENCES

Sergei Kucheyev, Ph.D.,

LLNL advisor/supervisor

Deputy group leader, Lawrence Livermore National Laboratory

kucheyev1@llnl.gov

Yan-Kai Tzeng, Ph.D.,

Prof. Steven Chu Group, Stanford University

Mentor and collaborator on LLNL–Stanford joint project

Associate scientist, SLAC-Stanford Battery Center

ytzeng@stanford.edu

Marie A. Charpagne, Ph.D.,

UIUC advisor

Assistant Professor, University of Illinois Urbana-Champaign

mcharp@illinois.edu

Leonardus Bimo Bayu Aji, Ph.D.,

LLNL mentor

Staff Scientist, Lawrence Livermore National Laboratory

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Leigh Winfrey, Ph.D.

Penn State Ph.D. advisor

Professor, Dean of Engineering at State University of New York – Maritime College,

lwinfrey@sunymaritime.edu